

Assignment 2 Theory (LOS)

Compare Permissions on Regular Files and Directories

Create a regular file named sample.txt and a directory named sample_dir. Create at least one file inside sample_dir.

Apply each of the following permissions using chmod. Perform the specified operations and record your observations.

Note: Perform the exercise as a normal user. Do not use sudo.

Permission	Regular File (sample.txt) - Test	Observation	Directory (sample_dir) - Test	Observation
400(r--)	Try to read and modify the file	The contents can be viewed, but editing is denied due to lack of write permission.	Try to list and enter the directory	The filename can be displayed, but the directory cannot be accessed or entered.
200(-w-)	Try to read and modify the file	The file cannot be read, but its contents can be modified.	Try to list, enter, create a file	Listing and entering are denied, so a new file cannot be created.
100(--x)	Try to read and execute the file	The file cannot be read or executed.	Try to list and enter the directory	The directory contents cannot be listed, but the directory can be entered.
500(r-x)	Try to read and modify the file	Reading is possible, whereas modification is refused.	Try to list and enter the directory	The directory can be listed and entered successfully.
600(rw-)	Try to read and modify the file	Both reading and modification are possible.	Try to list, enter and create a file	The filename can be listed, but entering the directory and creating files are not permitted.
700(rwx)	Try to read, modify and execute the file	Reading and modification are possible, and execution permission is also granted.	Try to list, enter, create and delete files	All the specified directory operations can be performed.

```

(praneetha@Praneetha)-[~/Desktop]
$ touch sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ mkdir sample_dir

(praneetha@Praneetha)-[~/Desktop]
$ touch sample_dir/file.txt

(praneetha@Praneetha)-[~/Desktop]
$ echo "Hello Linux" > sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ ls -l
total 40
drwxrwxr-x 2 praneetha praneetha 4096 Aug 13 18:37 assignment_1
-rw-rw-r-- 1 praneetha praneetha 0 Aug 18 23:13 file1.txt
-rw-rw-r-- 1 praneetha praneetha 0 Aug 18 23:05 file.txt
drwxrwxr-x 2 praneetha praneetha 4096 Aug 13 21:25 Linux
drwxrwxr-x 2 praneetha praneetha 4096 Aug 20 19:14 Logs
drwxrwxr-x 3 praneetha praneetha 4096 Aug 11 16:08 MSIS
drwxrwxr-x 2 praneetha praneetha 4096 Aug 19 22:23 practice
drwxrwxr-x 2 praneetha praneetha 4096 Aug 19 22:25 practice_2
drwxrwxr-x 2 praneetha praneetha 4096 Aug 19 22:32 practice_3
drwxrwxr-x 2 praneetha praneetha 4096 Aug 19 22:36 practice_4
drwxrwxr-x 2 praneetha praneetha 4096 Aug 22 22:33 sample_dir
-rw-rw-r-- 1 praneetha praneetha 12 Aug 22 22:34 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ ls -l sample_dir
total 0
-rw-rw-r-- 1 praneetha praneetha 0 Aug 22 22:33 file.txt

```

1. 400(r--)

```

(praneetha@Praneetha)-[~/Desktop]
$ chmod 400 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 400 sample_dir

```

chmod 400 sample.txt

```

(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Hello Linux

(praneetha@Praneetha)-[~/Desktop]
$ echo "Modified" > sample.txt
bash: sample.txt: Permission denied

```

chmod 400 sample_dir

```

(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
ls: cannot access 'sample_dir/file.txt': Permission denied
file.txt

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir
bash: cd: sample_dir: Permission denied

```

2. 200(-w-)

```
(praneetha@Praneetha)-[~/Desktop]
$ chmod 200 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 200 sample_dir
```

chmod 200 sample.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
cat: sample.txt: Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ echo "Modified" > sample.txt
```

chmod 200 sample_dir

```
(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir
bash: cd: sample_dir: Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ touch sample_dir/file1.txt
touch: cannot touch 'sample_dir/file1.txt': Permission denied
```

3. 100(--x)

```
(praneetha@Praneetha)-[~/Desktop]
$ chmod 100 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 100 sample_dir
```

Chmod 100 sample.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
cat: sample.txt: Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ ./sample.txt
/bin/bash: ./sample.txt: Permission denied
```

chmod 100 sample_dir

```
(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ cd ..
```

4. 500(r-x)

```
(praneetha@Praneetha)-[~/Desktop]
$ chmod 500 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 500 sample_dir
```

chmod 500 sample.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Modified

(praneetha@Praneetha)-[~/Desktop]
$ echo "Hi" > sample.txt
bash: sample.txt: Permission denied
```

chmod 500 sample_dir

```
(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
file.txt

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ cd ..
```

5. 600(rw-)

```
(praneetha@Praneetha)-[~/Desktop]
$ chmod 600 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 600 sample_dir
```

chmod 600 sample.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Modified

(praneetha@Praneetha)-[~/Desktop]
$ echo "Good morning" > sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Good morning
```

chmod 600 sample_dir

```
(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
ls: cannot access 'sample_dir/file.txt': Permission denied
file.txt

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir
bash: cd: sample_dir: Permission denied

(praneetha@Praneetha)-[~/Desktop]
$ touch sample_dir/file2.txt
touch: cannot touch 'sample_dir/file2.txt': Permission denied
```

6. 700(rwx)

```
(praneetha@Praneetha)-[~/Desktop]
$ chmod 700 sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ chmod 700 sample_dir
```

chmod 700 sample.txt

```
(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Good morning

(praneetha@Praneetha)-[~/Desktop]
$ echo "Good night" > sample.txt

(praneetha@Praneetha)-[~/Desktop]
$ cat sample.txt
Good night

(praneetha@Praneetha)-[~/Desktop]
$ ./sample.txt sample.txt
./sample.txt: line 1: Good: command not found

(praneetha@Praneetha)-[~/Desktop]
$ ls -l sample.txt
-rwx----- 1 praneetha praneetha 11 Aug 22 22:57 sample.txt
```

chmod 700 sample_dir

```
(praneetha@Praneetha)-[~/Desktop]
$ ls sample_dir
file.txt

(praneetha@Praneetha)-[~/Desktop]
$ cd sample_dir

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ touch new_file.txt

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ ls
file.txt  new_file.txt

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ rm new_file.txt

(praneetha@Praneetha)-[~/Desktop/sample_dir]
$ ls
file.txt
```

Questions

What does read (r) permission allow you to do with a regular file?

- Read permission allows the user to open, view, or copy the contents of a regular file.

What does read (r) permission allow you to do with a directory?

- For a directory, read permission allows the user to see the names of the files and folders stored inside it.

What does write (w) permission mean for a regular file?

- Write permission gives the user the ability to change, edit, or overwrite the contents of the file.

What does write (w) permission mean for a directory?

- Write permission on a directory permits operations such as creating, removing, or renaming files and folders within it.

What does execute (x) permission mean for a regular file?

- Execute permission allows a regular file to be run as a program or script, provided its contents are suitable for execution.

Why is execute (x) permission important for accessing a directory?

- Execute permission provides traversal access, allowing the user to enter the directory and access items within it when permitted.

Which permission combination allows the owner to fully access a regular file?

- The 700 (rwx-----) permission provides the owner with read, write, and execute permissions for the file.

Which permission combination allows the owner to fully access a directory?

- The 700 (rwx-----) permission gives the owner complete access to the directory, including listing, entering, creating, and deleting entries.

Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

- The effect of a permission depends on whether it is applied to a file or a directory. For a regular file, r is associated with viewing its contents, w allows its contents to be changed, and x allows it to be executed. For a directory, r allows its entries to be

listed, w permits changes to the entries, and x provides the ability to enter or traverse the directory. Therefore, the same permission symbols have different purposes depending on the type of filesystem object.